

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMPE604D - ROBOTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 How many degrees of freedom does an object in 2D space have?

a. 1	b. 2	c. 3	d. 6
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- 2 In Cartesian Configuration of Robotics, we get ----- shapes of work volume

a. Parallel Sphere	b. Cylindrical	c. Rectangular	d. Irregular
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- 3 Which kinematics is used to calculate the position of end-effector from some specified values?

a. Forward Kinematics	b. Reverse Kinematics	c. Inverse Kinematics	d. Backward Kinematics
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- 4 4/3 Direction Control Valve 3 represents

a. Position	b. Direction	c. Ways	d. All of the above
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- 5 Grippers are used to.....

a. Hold the objects	b. Sense the objects	c. Move the objects	d. Both (A) & (C)
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- 6 Which type of sensor is used to measure the distance between the vehicle and other objects in its environment:

a. Ultrasonic sensor	b. Tactile sensor	c. Motion sensor	d. None of these
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- 7 Which part of the robot provides motion to the manipulator and end-effector?

a. Actuator	b. Controller	c. Sensor	d. Processor
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- 8 A program language:

a. defines the form of the instruction	b. is always machine dependent	c. is never machine dependent	d. All of the above
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- 9 If the orientation changes without the change of position then the transformation is

a. Pure translation	b. pure rotation	c. Combined transformation	d. none of the above
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10 What is Artificial Intelligence?

a. Artificial Intelligence is a field that aims to make humans more intelligent	b. Artificial Intelligence is a field that aims to improve the security	c. Artificial Intelligence is a field that aims to develop intelligent machines	d. Artificial Intelligence is a field that aims to mine the data
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11 Which of the following is not an actuator?

a. Digital actuator	b. Hydraulic actuator	c. Pneumatic actuator	d. Electric actuator
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12 The kinematic part of manipulator is called

a. Links	b. Joints	c. End effectors	d. Sensors
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Q2. Solve the following.

- A) Explain with neat sketches various Manipulator and its role in Robot 6
- B) Explain in detail the forward and reverse kinematic and derive the equation for 2 DOF robot manipulator having link length L_1 , L_2 & Orientation angle θ_1 , θ_2 . 6

Q3. Solve the following.

- A) Explain with neat sketch various basic robot configurations with work envelope. Write applications of each configuration. 6
- B) What are different types of end effectors? Explain any two in detail. 6

Q4. Solve Any Two of the following.

- A) Explain working principles of dc motor. Write applications and key features. 6
- B) Describe construction and working of LVDT to measure the linear displacement. 6
- C) Explain robot programming methods. Write their limitations 6

Q5. Solve Any Two of the following.

- A) Write the selection criteria for Electric and Hydraulic drives. 6
- B) Explain the following (1) Position Sensors (2) Velocity Sensor (3) Proximity Sensor 6
- C) Explain application of robot in industry for assembly and inspection 6

Q6. Solve Any Two of the following.

- A) What is robot kinematics and dynamics? 6
- B) Explain the applications of AI in Robotics. 6
- C) Explain with neat diagram pneumatic drive used in Robotics. 6

*** End ***